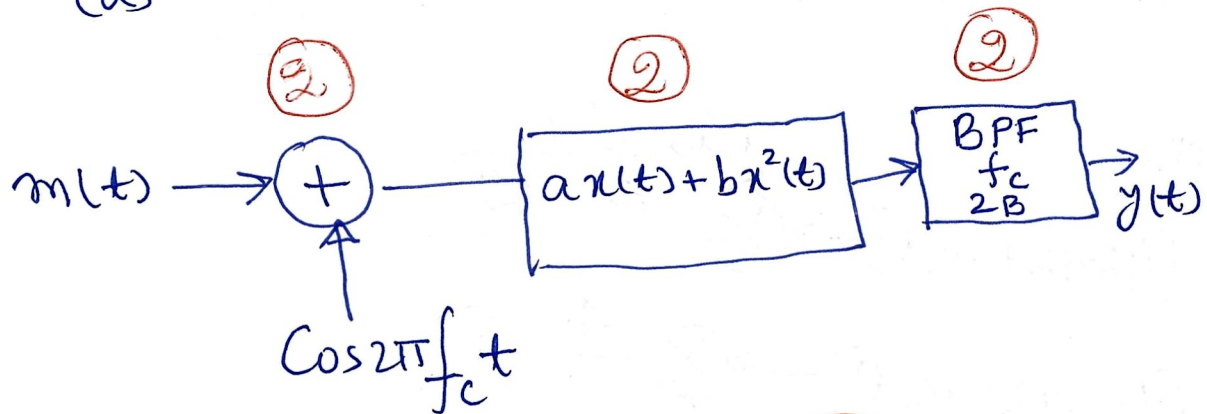


ECÉ/EEE F311 Communication Systems
Comprehensive Exam 2025-26
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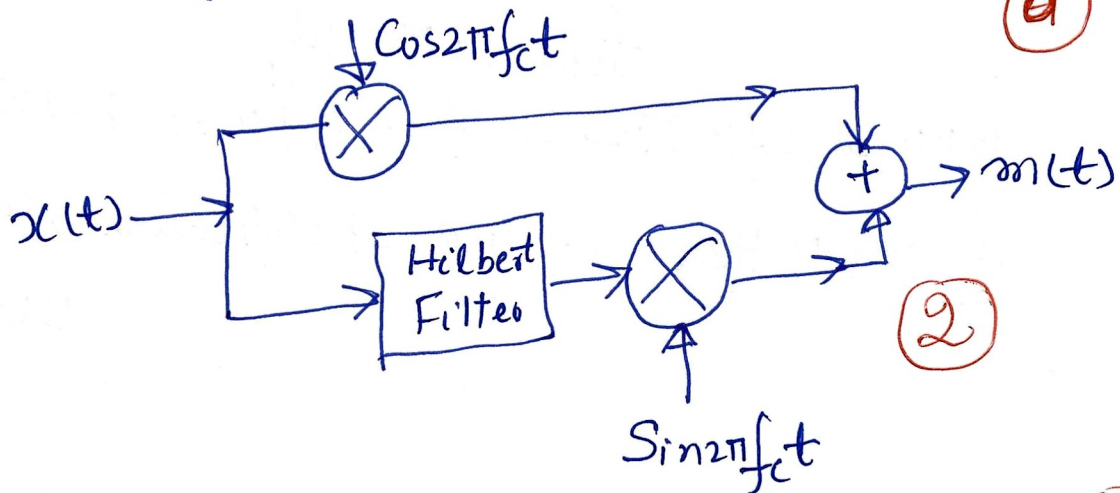
Q. 1 (a)



$$y(t) = [1 + m(t)] \cos 2\pi f_c t \quad (2)$$

Q. 1 (b)

$$m(t) = x(t) \cos 2\pi f_c t + x_h(t) \sin 2\pi f_c t \quad (6)$$



Q. 1 (c)

Signal power = 10 W (2) $B = 5 \text{ Hz}$ (2)

Noise power = 10^{-8} W (2)

SNR = 90 dB (2)

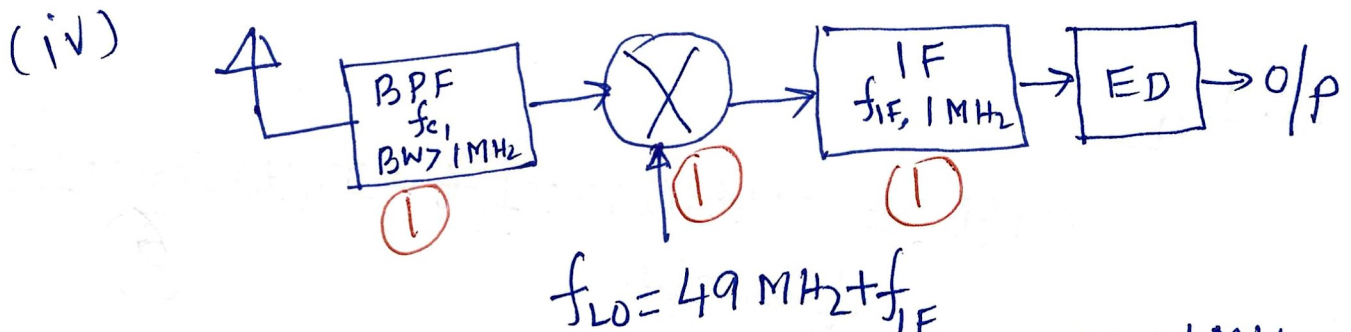
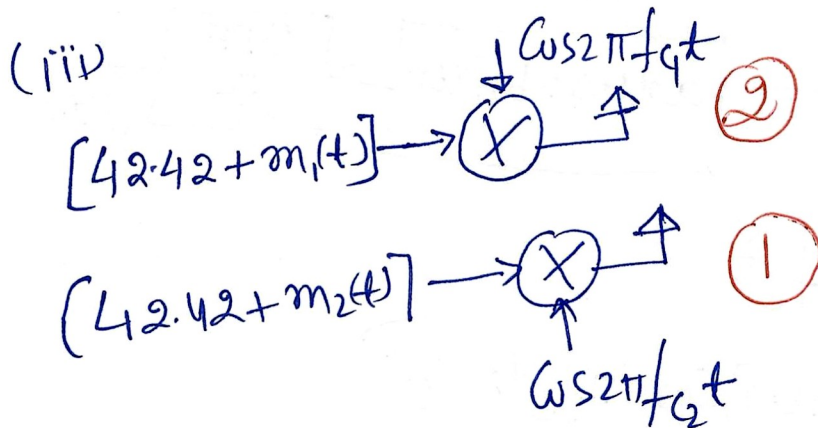
Q. 2

(a)

$$(i) \quad x_1(t) = (42.42 + m_1(t)) \cos 2\pi f_{c1} t, \quad f_{c1} = 49 \text{ MHz} \quad (1)$$

$$x_2(t) = (42.42 + m_2(t)) \cos 2\pi f_{c2} t, \quad f_{c2} = 51 \text{ MHz} \quad (2)$$

(ii) Transmitted power = 1000 Watt (3)



BPF specification $\Rightarrow f_{c1} = 49 \text{ MHz}, BW > 1 \text{ MHz}$

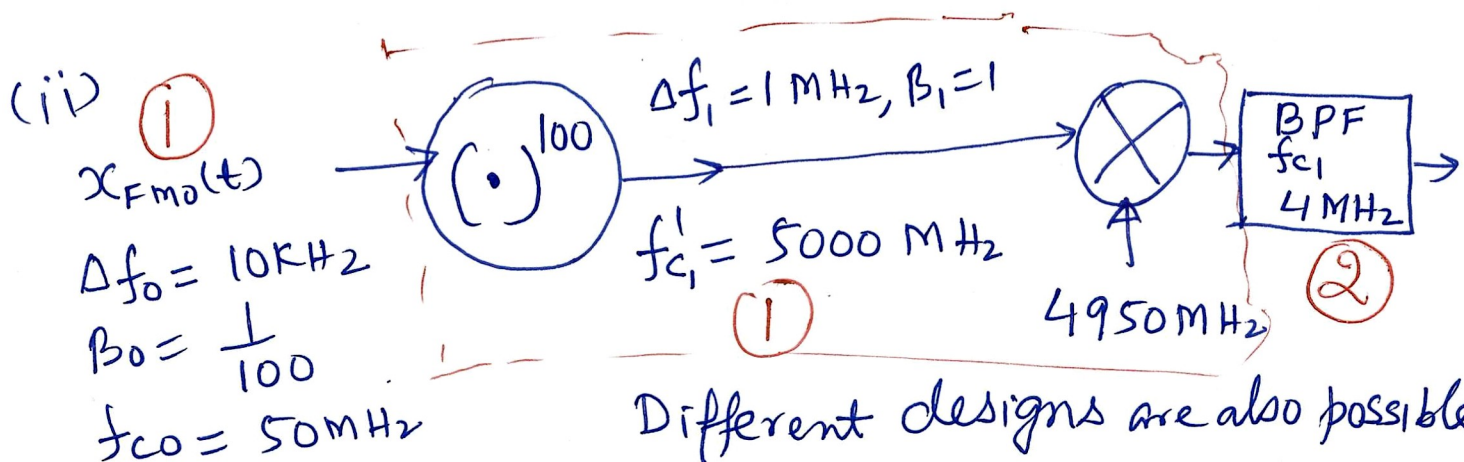
$f_{IF} > 1 \text{ MHz}$, IF BPF specification $\Rightarrow$ Center f_{IF}
 $BW = 1 \text{ MHz}$

Q.2 (b)

$$(i) \quad x_{FM_1}(t) = A_c \cos(2\pi f_{c_1} t + B_1 \sin 2\pi 10^6 t) \quad (1)$$

$$= A_c \cos(2\pi f_{c_1} t + 2\pi K_f \int_0^t m(\omega) dz)$$

$$\boxed{B_1 = 1 \text{ or } K_f = 1000} \quad (2) \quad \boxed{f_{c_1} = 50 \text{ MHz}} \quad (1)$$



$$(iii) \quad x_{FM_1}(t) = A_c \sum_{n=-\infty}^{\infty} J_n(1) \cos 2\pi(f_c + n f_m)t \quad (1)$$

Transmitted power

$$P_T = \frac{A_c^2}{2} \underbrace{|J_0(1)|^2}_{(1)} + \frac{A_c^2}{2} \underbrace{[2|J_1(1)|^2]}_{(1)} + \frac{A_c^2}{2} \underbrace{[2|J_2(1)|^2]}_{(1)} + \dots$$

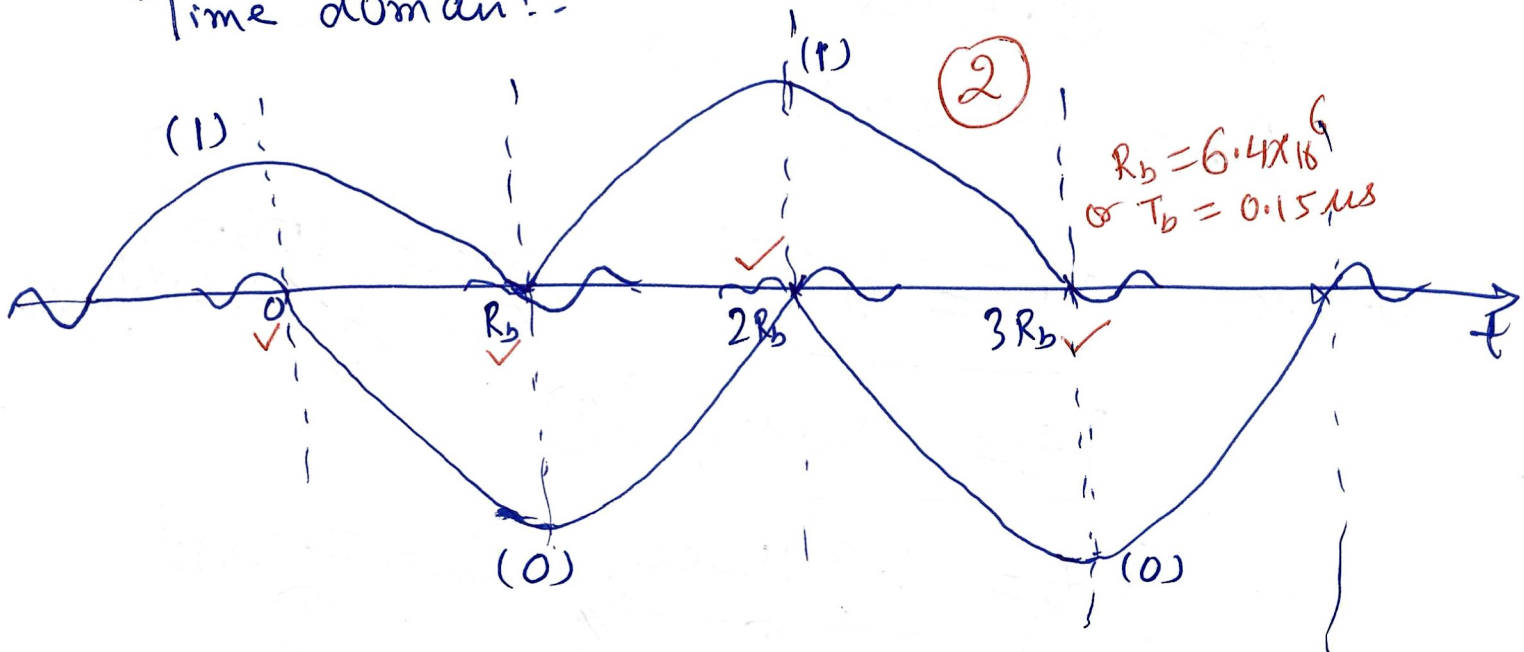
Q.3

(a) $B = 3.2 \text{ MHz}$, $b = 1.4 \times 10^9 \text{ bits}$

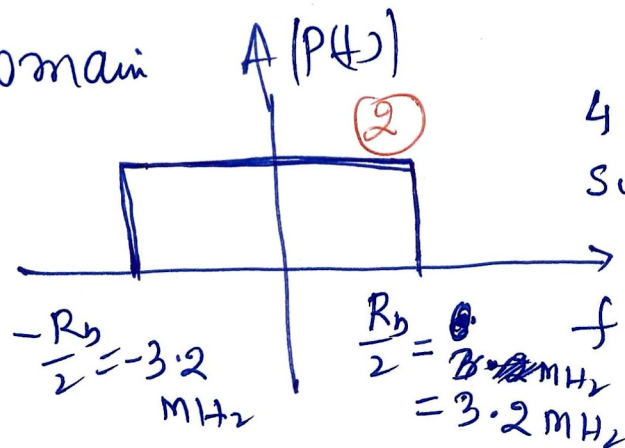
optimal pulse $p(t) = \text{Sinc}(\pi R_b t)$; $R_b = \underline{6.4 \text{ Mbps}}$

$P(f) = T_b \Pi\left(\frac{f}{R_b}\right)$; $T_b = 0.15 \mu\text{s}$

Time domain:-

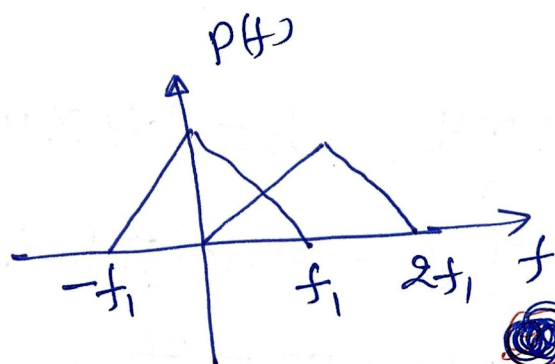


Frequency domain



4 rectangular pulses
superimposed.

Q.3 (b)



$$f_1 = 3.2 \text{ MHz}, \quad -f_1 = -3.2 \text{ MHz}$$

$$A = 0.31 \mu\text{s}$$

$$R_s = 3.2 \text{ Msps}$$

$$R_b = 19.2 \text{ Mbps}$$

Q.3 (c)

$$M = 8$$

$$t = 72.91 \text{ sec}$$

$$t_{\infty} = 1.944 \text{ sec}$$

$$\text{SNR} = 156.25$$

$$C_s = 3.64 \text{ bits/Symbol.}$$

$$C_{\infty} = 7.2 \times 10^8$$

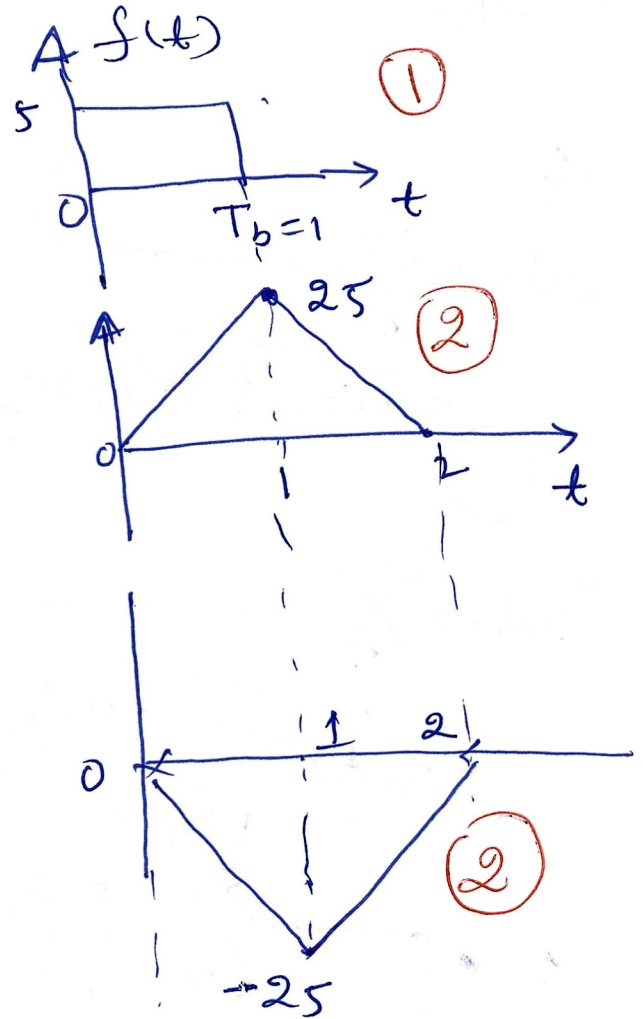
Q.4

(a)

Matched filter

O/P for $p(t)$

O/P for $q(t)$



Before MF

$P_s = 25W$, Noise power = $20W$

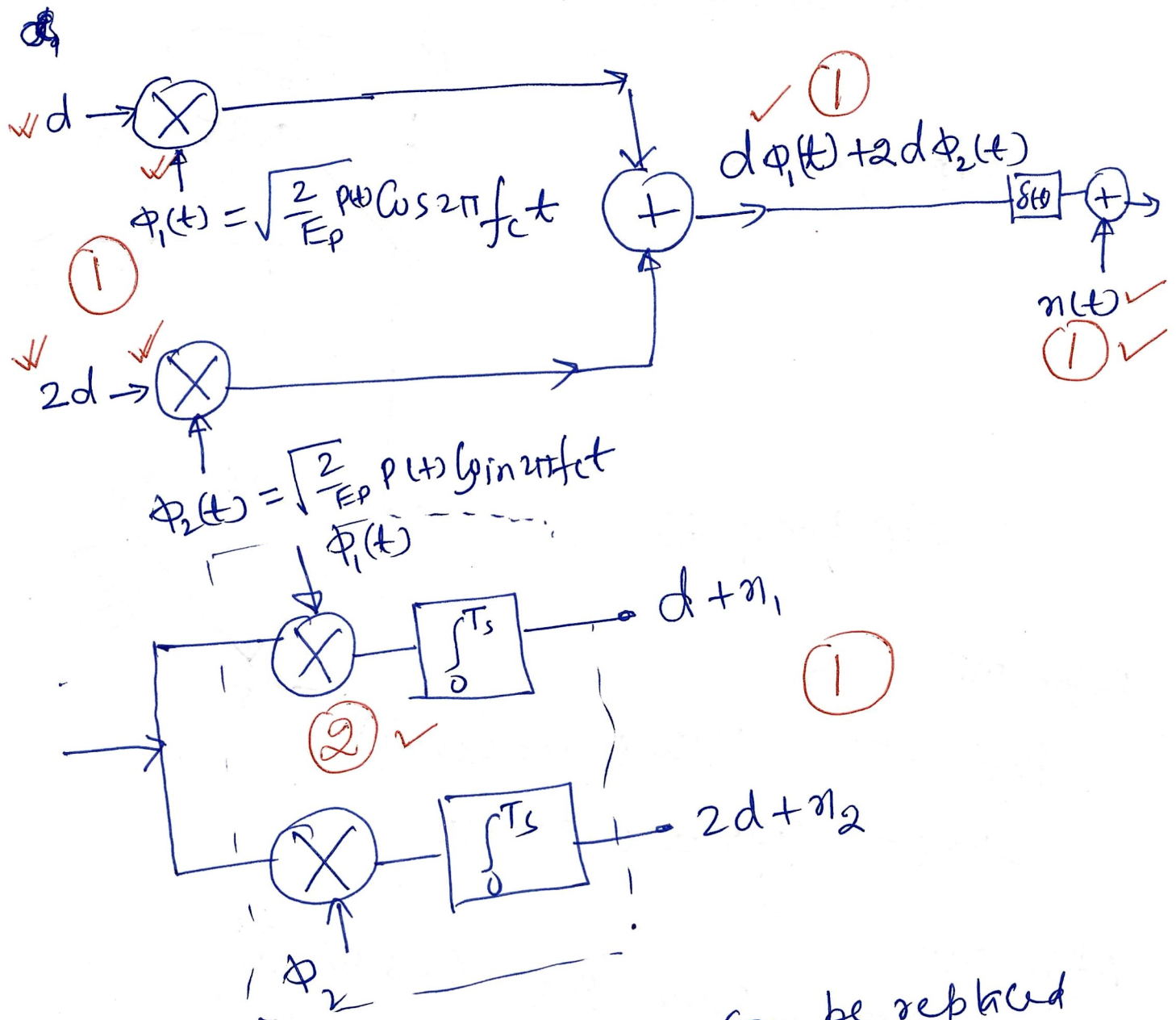
$SNR_i = 1.25 = 0.97 \text{ dB}$

After MF

Signal Power $P_s = 625W$
 Noise Power $N = 250W$

$SNR_o = 2.5 = 3.98 \text{ dB}$

Q. 4 (b)



The Correllator receiver can be replaced by the MF $\rightarrow \Phi_1(t - T_s)$
 $\rightarrow \Phi_2(t - T_s)$

$$P_c = \left(1 - 2Q \sqrt{\frac{6E_b}{5N_0}} \right)^2$$

Q. 5

(a) Kraft Inequality

$$\boxed{L_s = 1} \quad (6)$$

(b) Second order codes

0, 10, 110, 111

$$H(X) = 0.938 \quad (2)$$

$$\bar{L} = 1.29$$

$$\eta = 0.727 \quad (1)$$

3rd order codes

0, 100, 101, 110, 11100, 11101, 11110, 11111

$$H(X) = 1.407$$

$$\bar{L} = 1.598 \quad \eta = 0.88 \quad (2)$$

$$\boxed{\eta \rightarrow 1 \text{ if } \eta \rightarrow \infty} \quad (1)$$

eta

(c)

$$C_2 > C_1 > C_3$$

(4)

Justification: -

Cross-over decreasing in the order of increasing capacity

Max capacity

$$C_2 \rightarrow \frac{1}{2} \log_2 3 \text{ bits/symbol} \quad (2)$$

(d)

Block words
(4) $\left\{ \begin{array}{l} 000 \underline{0000}; \quad 001 \underline{0111}; \quad 010 \underline{1110}; \quad 011 \underline{1001} \\ 100 \underline{1011}; \quad 101 \underline{1100}; \quad 110 \underline{0101}; \quad 111 \underline{0010} \end{array} \right.$

$d_{\min} = 4$ Error detection = 3 bits (1)

Decoded bits: - 1001011 (1)